Factorization of Polynomials Over the Boolean Semifield

U-SOAR 2025 Summary

Micah Sherry

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Background

The Boolean semifield, denoted as $\mathbb{B}$, is an algebraic structure with two operations: logical OR, analogous to addition, and logical AND, analogous to multiplication. Our research explores the factorization of polynomials over the Boolean semifield, denoted as $\mathbb{B}[x]$.
A key challenge in $\mathbb{B}[x]$ is that factorization is not unique. Specifically, we investigate the factorization of a particular class of polynomials in $\mathbb{B}[x]$ namely:

$$f_n(x) = \sum_{i=0}^{n} x^i = x^n + x^{n-1} + \cdots + x + 1$$

Understanding the non-unique factorization of these polynomials is a central focus of our study. This research contributes to a deeper understanding of algebraic structures beyond traditional fields and rings.

Definitions

- A semifield is an algebraic structure with addition and multiplication. That does not have additive inverses.
- Let $\mathbb{B} = \{0, 1\}$ be the Boolean semifield equipped with logical OR as addition and Logical AND as Multiplication.
- Let $\mathbb{B}[x]$ be The set of all polynomials in x over $\mathbb{B}$.
- Let $f_n(x) \in \mathbb{B}$, be defined as $f_n(x) = x^n + x^{n-1} + \cdots + x + 1$.

Related Topics

1. Set Addition
   $A + B = \{a + b | a \in A \text{ and } b \in B\}$ where $A, B \subseteq \mathbb{N}$
2. Dismal Arithmetic (base $b \geq 2$) Let a and b be base-b numbers with digits in $\{0, 1, \ldots, b-1\}$.
   1. $a \oplus b$ is given by digitwise max (no carries).
   2. $a \otimes b$ is given by digitwise min, combined as in ordinary multiplication (no carries).
3. Numbral Arithmetic let a and b be binary sequences.
   1. $a + b$ is given by Bitwise OR.
   2. $a \times b$ is given by Shift-and-OR.
4. Polynomials in x over the Tropical semiring

Divisibility of $f_n(x)$

Lemma

Lemma 3.1 from the paper 'On the Sequence A079500 and Its Combinatorial Interpretations' [3] states conditions for divisibility in the numbral arithmetic: For each $n, d > 0$, it holds that $[d]$ is a divisor of $[2^n - 1]$ if and only if it ends with the digit 1, and it does not contain any subsequence of 0s having length greater than $n - k$, with k being the length of $[d]$.

Theorem

Let $g(x) \in \mathbb{B}[x]$ such that the constant term of $g(x)$ is 1. If $\deg(g(x)) = k$ and $2k - 1 \leq n$, then $g(x)$ divides $f_n(x)$.

Proof.

Let $\theta(h(x))$ be the isomorphism between $\mathbb{B}[x]$ and the numbral arithmetic $\theta(h(x)) = [h(2)]$. Assume $2k - 1 \leq n$ and let $d = \theta(g(x))$ (notice $[d]$ is length $k + 1$). Let m be the maximum length of zeros in $[d]$. Notice $m \leq k - 1$. Notice $\theta(f_n(x)) = 2^{n+1} - 1$. By the above lemma, d divides $2^{n+1} - 1$ because $k - 1 \leq n + 1 - (k + 1)$, which is equivalent to $2k - 1 \leq n$. And by applying θ^{-1} we see $g(x)$ divides $f_n(x)$. $\square$

Factorizations Of The Form $(x+1)h(x) = f_n(x)$

Theorem

Let G_n be the set of $g(x) \in \mathbb{B}[x]$ such that $g(x) = \sum_{i=1}^{n} a_i x^i$ and for all a_i if $a_i = 0$ then $a_{i+1} \neq 0$. Let $F(n)$ be the nth Fibonacci number. $|G_n| = F(n+2)$ for $n \geq 1$

Proof (by Strong induction).

$G_1 = \{x, 0x\}, \quad |G_1| = 2 = F(3) \quad \& \quad G_2 = \{x^2 + x,\ x^2,\ x\}, \quad |G_2| = 3 = F(4)$
Therefore the base cases hold. Assume that $|G_j| = F(j+2)$ for all $j \in \{1, 2, \ldots, k\}$ for some k. Let $g(x) \in G_{k+1}$. Consider the possible values for a_{k+1}.

1. Case 1. $a_{k+1} = 1$. In which case the rest of the polynomial can be any polynomial in G_k.
2. Case 2. $a_{k+1} = 0$ which implies $a_k = 1$. In which case the rest of the polynomial can be any polynomial in G_{k-1}.

So, $|G_{k+1}| = |G_k| + |G_{k-1}| = F_{k+2} + F_{k+1} = F_{k+3}$. Therefore, by the Principle of Mathematical Induction $|G_n| = F_{n+2}$. □

Factorizations Of The Form $(x+1)h(x) = f_n(x)$

Theorem

$(x+1)h(x) = f_n(x)$ if and only if $h(x) = x^{n-1} + g'(x) + 1$ where $g'(x) \in G_{n-2}$.

Proof.

($\rightarrow$): Assume that $(x+1)h(x) = f_n(x)$ and let $h(x) = \sum_{i=0}^{n-1} a_i x^i$. (Notice $a_{n-1} = a_0 = 1$ otherwise it would contradict the assumption). Now assume for the sake of contradiction that there exists $k < n-1$ such that $a_k = a_{k+1} = 0$. Consider the following:

$$(x+1)h(x) = x\sum_{i=0}^{n-1} a_i x^i + \sum_{i=0}^{n-1} a_i x^i.$$

Notice the $k+1$ term will have coefficient $a_k + a_{k+1} = 0$, contradicting the original assumption.
($\leftarrow$): Assume $h(x) = \sum_{i=0}^{n-1} a_i x^i$ with $a_{n-1} = a_0 = 1$ and for all a_i, if $a_i = 0$ then $a_{i+1} \neq 0$. Consider the following:

$$(x+1)h(x) = \sum_{i=0}^{n-1} a_i x^{i+1} + \sum_{i=0}^{n-1} a_i x^i = a_{n-1}x^n + \sum_{i=1}^{n-1}(a_{i-1} + a_i)x^i + a_0 = f_n(x).$$

Therefore, the theorem holds. $\square$

Factorizations Of The Form $(x+1)h(x) = f_n(x)$

Theorem

There are $F(n)$ polynomials $h(x)$ such that $(x+1)h(x) = f_n(x)$.

Proof.

Consider the following cases:

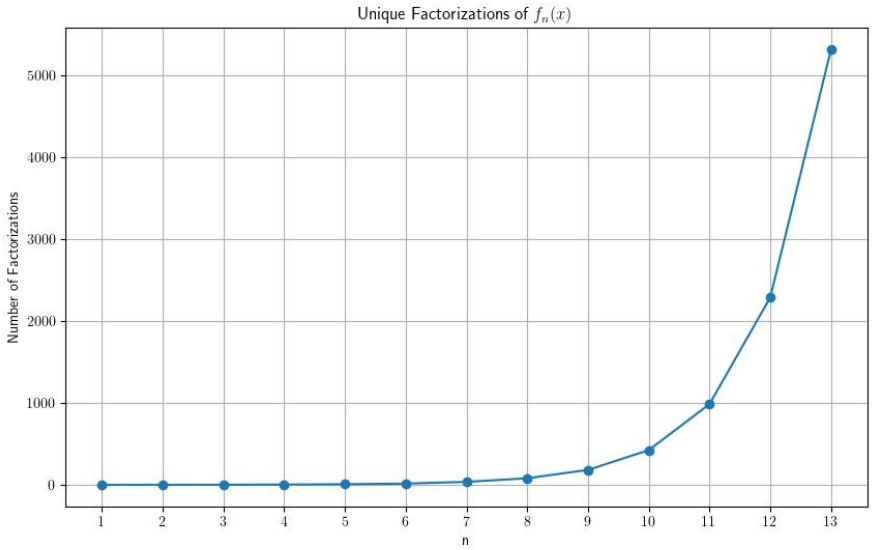
- $n = 1$. $h(x) = 1$ is the only valid $h(x)$ such that $(x+1)h(x) = f_1(x)$. So, the theorem holds in this case.
- $n = 2$. $h(x) = x + 1$ is the only valid $h(x)$ such that $(x+1)h(x) = f_2(x)$. So, the theorem holds in this case.
- $n \geq 3$. In this case, the theorem follows directly from the two previous theorems.

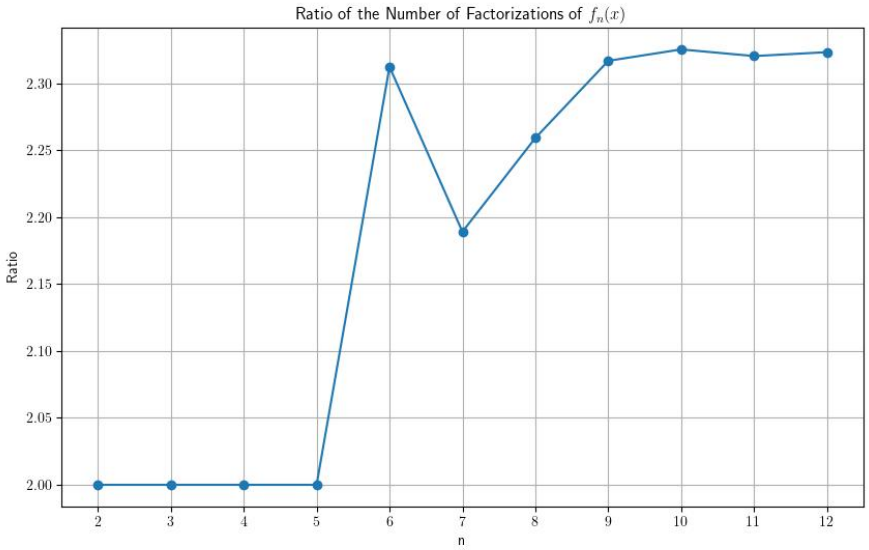
$\square$

Unique factorization Examples

n	$f_n(x)$	Factorizations
2	$x^2 + x + 1$	$(x+1)^2$
3	$x^3 + x^2 + x + 1$	$(x+1)^3$ $(x+1)(x^2+1)$
4	$x^4 + x^3 + x^2 + x + 1$	$(x+1)^4$ $(x+1)^2(x^2+1)$ $(x^3+x+1)(x+1)$ $(x+1)(x^3+x^2+1)$
5	$x^5 + x^4 + x^3 + x^2 + x + 1$	$(x+1)^5$ $(x+1)^3(x^2+1)$ $(x^3+x+1)(x+1)^2$ $(x+1)^2(x^3+x^2+1)$ $(x+1)(x^2+1)^2$ $(x+1)(x^4+x^2+x+1)$ $(x^3+1)(x+1)^2$ $(x+1)(x^4+x^3+x^2+1)$

Growth Rate of Factors





Current Work

1. Establishing $\prod_{i=1}^{n} Fibb(i)$ as a lower bound of number of unique factors.
2. Determining the factors of the form $(x^k + 1)h(x) = f_n(x)$.

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